

# The Earth and Mars: Atmospheric Evolution

Target: Grades up to A / A\*

Topic B6: Global Challenges · Extended Response Assessment

Candidate Name: ..... Class / Set: ..... Date: .....

## 1 Atmospheric composition and planetary evolution

The early atmospheres on Earth and on Mars were similar.  
They **both** contained mainly **carbon dioxide** and **water vapour**.

The atmospheres on the two planets are now very different.

The table shows the composition of the atmosphere on Mars now.

| Gas            | Composition now (%) |
|----------------|---------------------|
| carbon dioxide | 95                  |
| oxygen         | traces              |
| water vapour   | traces              |
| other gases    | 4                   |

The average surface temperature of Mars is now  $-55^{\circ}\text{C}$ .

Use the information to describe how the atmosphere on Mars has changed. Compare these changes to what has happened to the atmosphere on Earth. Give reasons for the changes to the Earth's atmosphere.

 In your answer, you should compare changes on both planets and give scientific reasons for changes on Earth. [QER · 6 Marks]

**[6 Marks]**

## Official Mark Scheme: Atmospheric Evolution

Use L1, L2, L3 in Scoris

Scoris Level-of-Response Assessment Criteria (Grades up to A / A\*)

## OFFICIAL EXAM BOARD KEY

This question is targeted at grades up to A / A\*

Indicative scientific points may include:

- Percentage carbon dioxide has increased on Mars
- Percentage water vapour has decreased on Mars
- Some oxygen has appeared on Mars
- Percentage carbon dioxide has decreased on Earth
- Percentage water vapour has decreased on Earth
- Percentage oxygen has increased on Earth
- Water vapour has disappeared from both the atmosphere of Earth and Mars by condensation or freezing
- Oceans have formed
- Carbon dioxide has disappeared from Earth because it has dissolved in oceans
- Plants have added oxygen
- Plants have removed carbon dioxide from Earth's atmosphere
- Lowering of amounts of carbon dioxide has given much higher proportion of nitrogen on Earth

accept: atmosphere on Mars very small / thin compared to that on Earth.

Use the L1, L2, L3 annotations in Scoris; do not use ticks.

## Indicative Scientific Points (Examiner Guidance)

- **Mars changes:** Percentage CO<sub>2</sub> increased (now 95%); water vapour decreased to trace amounts; some oxygen appeared (traces).
- **Water vapour removal:** Water vapour disappeared from both Earth and Mars by condensation or freezing (Mars average -55 °C).
- **Earth changes:** Percentage CO<sub>2</sub> decreased drastically (from ~95% to 0.04%); water vapour decreased; oxygen increased (to 21%).
- **Oceans formed:** Water vapour condensed to form oceans as Earth cooled below 100 °C.
- **CO<sub>2</sub> dissolved:** Carbon dioxide dissolved in oceans and formed insoluble carbonate precipitates/sedimentary rocks (e.g., limestone).
- **Biological terraforming:** Algae and plants evolved; carried out photosynthesis, adding oxygen and removing CO<sub>2</sub>.
- **Nitrogen accumulation:** Lowering CO<sub>2</sub> and inertness of N<sub>2</sub> gave a much higher proportion of nitrogen (78%).

**Special Examiner Acceptance:** Accept statement that atmosphere on Mars is *very small / thin* compared to that on Earth.

| Level   | Marks | Description of Student Performance                                                                                                                                                                                                                                                                                                             |
|---------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 3 | 5–6   | <b>Comprehensive &amp; Detailed:</b> Compares atmospheric changes on <i>both Mars and Earth</i> , and provides clear, correct scientific reasons for changes to Earth's atmosphere (oceans condensing, CO <sub>2</sub> dissolving/rocks, photosynthesis). Coherent, well-structured scientific argument with specialist terms used accurately. |
| Level 2 | 3–4   | <b>Sound Understanding:</b> Describes changes to both atmospheres with at least one correct scientific reason given for Earth, OR explains reasons for changes on Earth with limited comparison to Mars. Communication is clear with some relevant scientific terminology.                                                                     |
| Level 1 | 1–2   | <b>Basic / Limited:</b> States one or two changes to the atmosphere of Mars or Earth (e.g., CO <sub>2</sub> decreased on Earth / increased on Mars / oxygen increased) without linking to reasons. Fragmented reasoning.                                                                                                                       |
| Level 0 | 0     | No creditworthy response.                                                                                                                                                                                                                                                                                                                      |

**Scoris Annotation Instruction:** Use the **L1, L2, L3** annotations in Scoris; **do not use ticks**. Award 6 marks for full comparison + mechanisms (oceans, carbonate rocks, photosynthesis, and temperature/freezing).